

INDIAN STATISTICAL INSTITUTE

Practice Problems, August 20, 2026
M.TECH(CS/CrS) I YEAR

INTRODUCTION TO PROGRAMMING

1. Write a C program to evaluate a postfix expression using a stack implemented with a dynamically resizable array. The input expression will contain integer operands and the binary operators $+$ and $*$ representing addition and multiplication respectively. More specifically, the input will consist of a sequence of whitespace-separated tokens representing a postfix expression. Each token would be an integer (positive or negative) or an operator $+$ or $*$. The input expression would end with EOF. Some example valid inputs are as follows:

2 3 + 4 *

23 32 + -15 60 + *

2 3 + 41 5000 + *

Note that the above expressions represent the infix expressions $(2 + 3) * 4$, $(23 + 32) * (-15 + 60)$, and $(2 + 3) * (41 + 5000)$ respectively.

For a valid input your program should output an integer which would be the evaluation of the given input expression. For invalid inputs, your program should be able to report that the input is invalid.